



# Gender Differences in Time Spent on Education

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## Introduction/Background

- Studies and news articles have highlighted how boys are falling behind girls in school with
  - lower reading scores (New York Times)
  - a graduation rate of 83% for boys and 89% for girls in 2023 (Brookings)
  - 66% women enrolled in college compared to 57% of men in 2022 (National Center for Education Statistics).

To investigate if this trend manifests itself in how much time students spend on education, I seek to answer the following questions:

**Has time spent on educational activities varied by gender from 2003-2024? Are there other factors, including income, location, race or their interactions with gender that contribute to differences in time spent on education?**

## Data

### Data Sources

- Roster, Respondent, Activity, and Current Population Survey files from the American Time Use Survey Data (ATUS) spanning 2003-2024.

### Sample

- 16,898 full-time students aged 15-24 in the ATUS dataset

### Data Cleaning

- To define educational activities, I included all categories under the major category of education with the label 06. The main activities include taking class, extracurricular club activities excluding sports, and homework.
- I summed all the time people spent on educational activities in a day, and filled in 0's for people who had no recorded time spent on education.
- I then aggregated several categorical variables into broader categories to avoid having too many levels, including income to 5 levels, race to 6 levels, and year into 4 periods of 4 years each with the last period being 6 years.

## Models

### Predictors

- Primary Predictor: Gender
- Secondary Predictors: Household Family Income, Race, Region
- Interactions w/ gender: Race, Income, Region, Year, College vs. High School
- Other predictors to control for: Weekday vs. Weekend, Month, Period, College vs. High School

- 4 models: 2 logistic regression models and 2 linear regression models**
- Each model type is run twice with:
  - A focus on gender differences with the variable period (aggregated years).
  - An interaction between numeric year and gender to see if there is change over time by gender.

- Logistic Regression Model Outcome:** Whether or not the student spent any time on educational activities at all
- Linear Regression Model Outcome:** For students who spent any time on education (n = 9024), the logged minutes spent on educational activities

## Visualizations

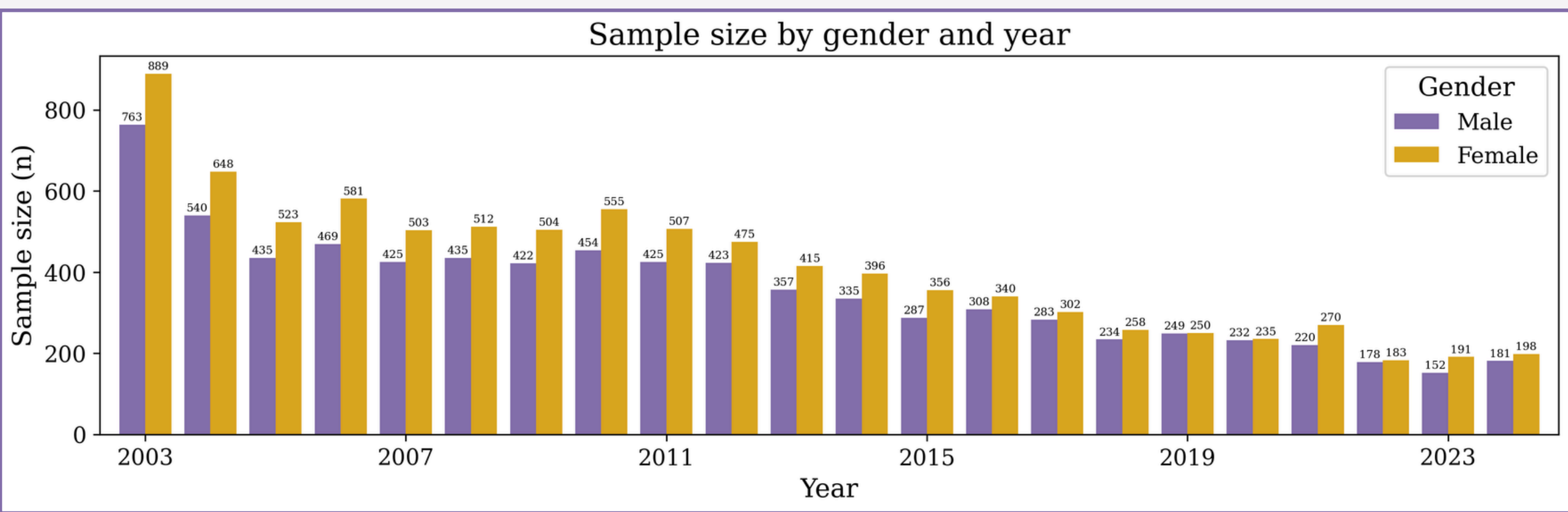


Figure 1: Sample size decreasing over time.

## Results

| ANOVA P-Values            | Model 1: Logistic Regression with Periods | Model 2: Logistic Regression with Numeric Year | Model 3: Logistic Regression with Periods | Model 4: Logistic Regression with Numeric Year |
|---------------------------|-------------------------------------------|------------------------------------------------|-------------------------------------------|------------------------------------------------|
| Gender                    | Uninterpretable                           | 0.677                                          | Uninterpretable                           | 0.602                                          |
| Year                      |                                           | 0.196                                          |                                           | 0.542                                          |
| Period (aggregated years) | < 0.0001                                  |                                                | 0.846                                     |                                                |
| Race                      | Uninterpretable                           | Uninterpretable                                | Uninterpretable                           | Uninterpretable                                |
| Family Income             | Uninterpretable                           | Uninterpretable                                | Uninterpretable                           | Uninterpretable                                |
| Region                    | Uninterpretable                           | Uninterpretable                                | Uninterpretable                           | Uninterpretable                                |
| College v. High School    | Uninterpretable                           | Uninterpretable                                | Uninterpretable                           | Uninterpretable                                |
| Weekend/ Weekday          | < 0.0001                                  | < 0.0001                                       | < 0.0001                                  | < 0.0001                                       |
| Month                     | < 0.0001                                  | < 0.0001                                       | < 0.0001                                  | < 0.0001                                       |
| Gender x Year             |                                           | 0.699                                          |                                           | 0.612                                          |
| Gender x Race             | 0.843                                     | 0.839                                          | 0.336                                     |                                                |
| Gender x Income           | 0.446                                     | 0.512                                          | 0.191                                     | 0.227                                          |
| Gender x Region           | 0.351                                     | 0.353                                          | 0.53                                      | 0.325                                          |
| Gender x School Level     | 0.016                                     | 0.021                                          | 0.032                                     | 0.028                                          |

| ANOVA P-Values                        | Model 1: Logistic Regression with Periods | Model 2: Logistic Regression with Numeric Year | Model 3: Logistic Regression with Periods | Model 4: Logistic Regression with Numeric Year |
|---------------------------------------|-------------------------------------------|------------------------------------------------|-------------------------------------------|------------------------------------------------|
| Gender + interactions with Gender     | < 0.0001                                  | < 0.0001                                       | 0.027                                     | 0.038                                          |
| Year (numeric) + interactions         |                                           | < 0.0001                                       |                                           | 0.662                                          |
| Race + interactions                   | < 0.0001                                  | < 0.0001                                       | < 0.0001                                  | < 0.0001                                       |
| Race + interactions                   | < 0.0001                                  | < 0.0001                                       | 0.02                                      | 0.023                                          |
| Region + interactions                 | 0.008                                     | 0.007                                          | 0.53                                      | 0.53                                           |
| College v. High School + interactions | < 0.0001                                  | < 0.0001                                       | < 0.0001                                  | < 0.0001                                       |

| Logistic Regression- Variables of Interest | Odds Ratio | P-Value  | 95% Confidence Interval |
|--------------------------------------------|------------|----------|-------------------------|
| Female                                     | 1.435      | 0.003    | (1.127, 1.828)          |
| Asian                                      | 1.722      | < 0.0001 | (1.394, 2.133)          |
| Female x College                           | 0.845      | 0.016    | (0.74, 0.97)            |
| 2008-2012                                  | 1.12       | 0.009    | (1.028, 1.219)          |
| 2008-2012                                  | 1.215      | < 0.0001 | (1.105, 1.337)          |
| 2008-2012                                  | 1.165      | 0.003    | (1.054, 1.288)          |

| Linear Regression | Coefficient Estimate | P-Value  | 95% Confidence Interval |
|-------------------|----------------------|----------|-------------------------|
| Female            | -8.04                | 0.145    | (-17.66, 2.71)          |
| Asian             | 20.78                | < 0.0001 | (10.69, 31.79)          |
| Weekend           | -53.98               | < 0.0001 | (-55.65, -52.24)        |
| Female x Hispanic | 9.83                 | 0.04     | (0.29, 20.29)           |

Multiple R-squared: 0.21

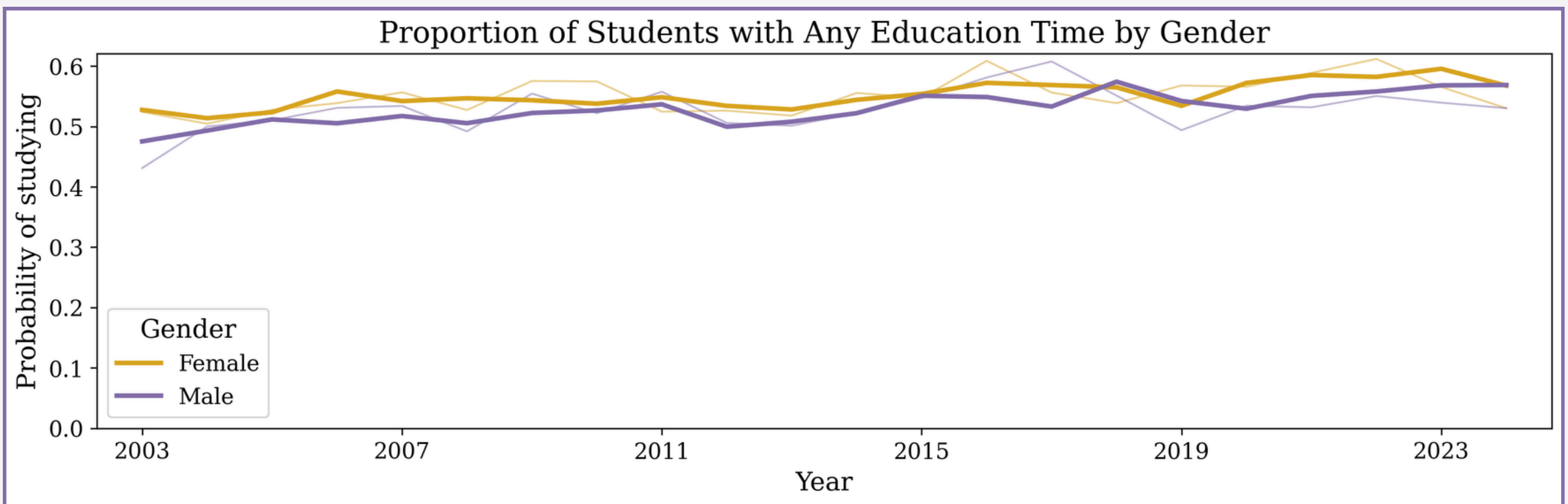


Figure 2: Probability of studying at all is statistically significantly higher for women.

Note: Lighter lines = actual proportions and darker lines = predicted probabilities

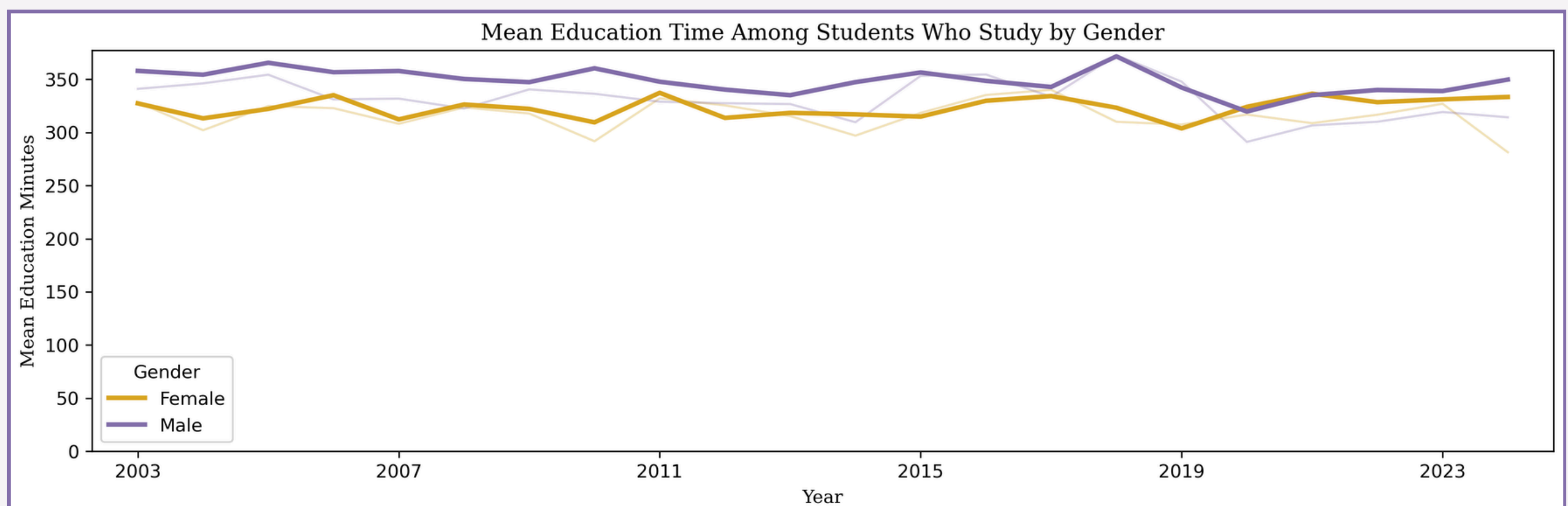


Figure 3: No statistically significant difference between men and women in amount of time studied if they do study.

Note: Lighter lines = actual means and darker lines = predicted means

## Discussion and Conclusion

- There is a **statistically significant difference between boys and girls in whether or not they spend time on education at all**. Girls have a 43.5% higher overall odds of studying on their survey day compared to boys.
- However, there is no statistically significant gender gap in how much time students spend on educational activities if they did engage in them.

### Limitations

- ATUS sample size is decreasing across years, which possibly affects the year variation change in education time.
- Near equal amount of weekends and weekdays in the ATUS dataset- weekend overrepresentation and less time spent on educational activities.
- Family Income is grouped into fixed buckets
- Mild heteroskedasticity for the linear regression model, as there is a lot of variability in educational time across levels of independent variables
  - Handled by HC3, a type of heteroskedasticity consistent robust standard errors, which manage high leverage observations and is used for small- medium sample sizes

### Future work

- Disaggregate Asian by specific race
- Explore other interactions, such as interactions for race and income

## References

- U.S. Bureau of Labor Statistics. (2024). American Time Use Survey (ATUS). <https://www.bls.gov/tus/>
- Miller, C. C. (2025, May 13). It's not just a feeling: Data shows boys and young men are falling behind. The New York Times. <https://www.nytimes.com/2025/05/13/upshot/boys-falling-behind-data.html>
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- National Center for Education Statistics. (2023). Digest of education statistics 2023, Table 302.10: Number of recent high school completers and percent enrolled in college, by sex and level of institution (1960-2022). U.S. Department of Education. [https://nces.ed.gov/programs/digest/d23/tables/dt23\\_302.10.asp](https://nces.ed.gov/programs/digest/d23/tables/dt23_302.10.asp)